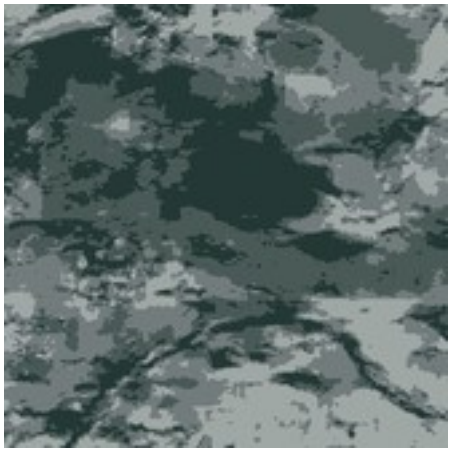
SAR Image Analysis Report

Original SAR (not found; preview shown)

Colorized



K-Means Segmented

Metadata:

- Original File: Not available
- Colorized File: colorized_1763464970.png
- Processed On: 2025-11-18 16:53:02 IST
- SAR Classifier Confidence: 0.000
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Water (Blue Tone) | Mean RGB: (115, 124, 124) | Area: 26.78%

Cluster 1: Likely Vegetation (Green Tone) | Mean RGB: (160, 168, 166) | Area: 17.75%

Cluster 2: Likely Vegetation (Green Tone) | Mean RGB: (35, 56, 52) | Area: 22.56%

Cluster 3: Likely Vegetation (Green Tone) | Mean RGB: (73, 89, 86) | Area: 32.92%

Land Cover Summary:

- Likely Water (Blue Tone) → 26.78%
- Likely Vegetation (Green Tone) → 17.75%
- Likely Vegetation (Green Tone) → 22.56%
- Likely Vegetation (Green Tone) → 32.92%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely vegetation (green tone) (73.23% combined).

Secondary presence: likely water (blue tone) (26.78%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is vegetation (32.78%). Strong vegetation signal suggests this area is suitable for agricultural monitoring — consider crop health and biomass monitoring. The second-largest class is water (26.6%), so combined management strategies addressing both vegetation and water are advisable. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.000
Dominant Class Percentage: 32.92%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.